Homework #3

Points: 15

Deadlines:
Groups 1, 2, 3, and **4**: **22 November** 2026, 23:59
Group 5: **24 November** 2026, 23:59

The main aim of this homework is to assess your capabilities to develop useful **Vue +VueX-based apps.** You are required to create your app using **Vue +Vuex techniques.**

Tasks

1. Your App should have two different "pages", one of them should be the **main "page"** that contains **at least 10 posts**, and the other "page" should be the **signup "page"**. **(0 points, but it is mandatory for completing the other tasks).**
   1.1. The **main page** should have the structure shown in Figure 1.

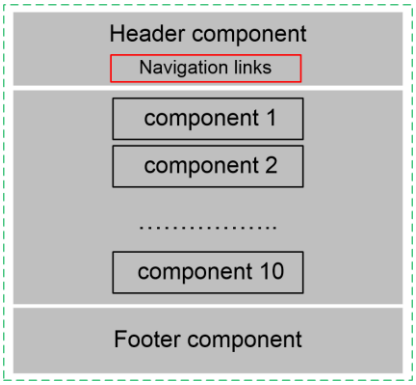


Figure 1: The main page structure

   1.2. The **signup page** should have the structure shown in Figure 2.

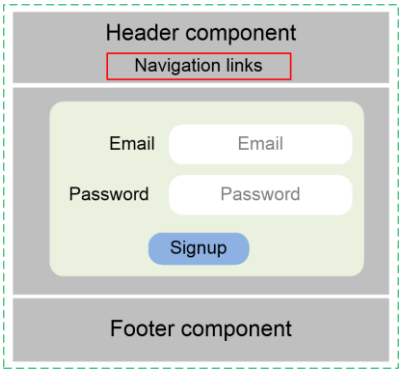


Figure 2: The signup page structure

2. The **header** should be a **component** that appears on both "pages" of your App, and it should include navigation links/routes to both pages **(2 points).**
3. The **footer** should also be a **component** that appears on both "pages" of your app, and it should include some basic information that traditional footers include (e.g., copyright, etc.) **(1 point).**
4. Concerning the main page:
   4.1. **A post** should have the same structure as the previous homework, but it has to be a **component** that relies on the **Vuex global store** to acquire/fetch its content information **(3 points).**
   4.2. **A post** should have a **"like" button** that is accompanied by the **number of likes**, which increases when we press the button **(2 points).**
   4.3. On the main "page", you need to add a **button** that resets the **number of likes** in all posts to zero **(2 points).**

To avoid any confusion, the main page should look like Figure 3.



Figure 3. The expected appearance of the main page

5. The **signup** "page" should include a form that contains the elements shown in Figure 2 (**1 point**).
6. The **signup** "page" should provide a password validation that checks for the **following conditions**:
   - The password should be of a specific length (at least 8 characters and less than 15 characters).
   - Includes at least one uppercase alphabet character.
   - Includes at least two lowercase alphabet characters.
   - Includes at least one numeric value.
   - It should start with an uppercase letter.
   - It should include the character "_"

When the password does not satisfy one or more of the previous conditions, a text should appear within the form stating that the **password is not valid**, **followed by the condition(s) that have not been satisfied**. For example, "the password is not valid - Includes at least one numeric value". **(3 points)**

7. The styling of your overall App **(1 point).**

Note: the term "page" has been used while describing the tasks, yet it might mean a **view** or a **single page component (SPC)**; you should know which one is better to use.

Rules for homework submission and discussion

1. Through Moodle, submit a text file (*.txt) that contains your **Team code**, Name(s), and a **valid and accessible link to the repository** that contains your **homework**. You can make your repo **private,** but you need to add your teacher and me as collaborators. Still, you need to submit the link to it through Moodle. **Note:** if the **link** to your **repository** is **not accessible** or **valid** for any reason, you might not be allowed to discuss your homework or at least you will lose **5 points**.
2. You are **not allowed to modify** the content of your repo **after the deadline.**
3. You are **not allowed** to share the link to your repository with anyone **except your lab teacher and the course lecturer**.
4. You **must send** the link to your repository to your teacher via a **direct message** in **Slack, and include** the **team number** and name(s) in the message.
5. Your lab teacher will share a link to an Excel sheet containing 10-minute discussion slots. Each team must book one of the available slots, join the discussion at the scheduled time, and be prepared to present their work. Make sure your project is up and running before the discussion begins.
6. **All team members should attend the discussion** of their homework; you **will not be allowed to discuss** if your team **is not complete**. If you already know that your team will not be complete because one or more of the members cannot attend due to another commitment, **contact me** as soon as possible, and we can find a solution.
7. You have to submit your homework by the defined deadline, and **you will lose 0.5 point for each hour of delay.**

AI-Assisted Coding: Students may use AI-assisted coding tools when completing the homework. However, **you are fully responsible for the code you submit**. You should understand how your code works and be able to explain and answer questions about **any part of your submitted code**.